

Assignment: Solar Shadow Analysis Web Application

Objective

Develop a web-based 3D application that simulates a solar installation site and analyzes the impact of surrounding structures on solar panel performance.

Functional Requirements

1. 3D Scene

Create an interactive 3D scene containing:

- Two solar tables
- Buildings represented as cuboids
- Water tanks represented as cylinders

Each solar table should:

- Contain a 2×3 panel arrangement (6 panels total)
- Have a fixed tilt angle of 15 degrees
- Be rendered as a single table structure composed of individual panels

2. Object Manipulation

Provide controls that allow users to modify or add objects within the scene in real time.

Buildings

Users should be able to modify:

- Position (X, Y)
- Width
- Length
- Height

Water Tanks

Users should be able to modify:

- Position (X, Y)
- Radius

- Height

Solar Tables

Users should be able to:

- Move the entire table within the scene
- Adjust its position (X, Y)

The 2×3 panel arrangement and 15-degree tilt should remain intact while the table is moved.

All changes should be reflected immediately in the rendered scene.

3. Sun Position Simulation

Implement a sun simulation system.

Users should be able to:

- Adjust the sun position manually using azimuth and elevation controls, or
- Specify a date and time and have the sun position computed automatically.

Examples:

- 21 June, 9:00 AM
- 21 June, 1:00 PM
- 21 December, 4:00 PM

The application should dynamically update shadows based on the selected time.

Candidates may use any reasonable approach for solar position calculation and should document their methodology and assumptions in the README.

Bonus:

- Support playback of sun movement throughout the day.
- Provide shadow analysis for multiple time intervals.
- Display summary metrics across a selected date or time range.

4. Shadow Analysis

Analyze the shadows cast by buildings and tanks onto the solar tables.

The application should:

- Visualize shadows in the 3D scene
- Determine the amount of shading affecting each panel
- Provide quantitative metrics describing the shadow impact
- Present the analysis results in a clear and intuitive manner

The methodology used for shadow analysis is left to the candidate and should be documented.

5. Efficiency Analysis

Based on the computed shadow coverage, provide an approach to estimate the efficiency impact on solar panels.

The objective is not simply to determine whether a panel is shaded, but to reason about how shading affects energy production.

Consider questions such as:

- How would you quantify the amount of shadow falling on a panel?
- Is total shaded area sufficient, or does the location of the shadow on the panel matter?
- How would you compare two panels where the same shaded area is distributed differently?
- How would you aggregate results across all panels in a table?
- How would you classify or score panel performance based on shading conditions?

Candidates are expected to propose and justify their own methodology for efficiency analysis and panel classification.

A brief explanation of the chosen approach, assumptions, and limitations should be included in the README.

Expected Output

Submit a ZIP file containing:

Source Code

Complete source code required to run the application locally.

Deployed Application

A publicly accessible deployment of the application (through vercel/netlify/any equivalent hosting platform)

README.md

The README should include:

- Setup and run instructions

- Project architecture overview
- Assumptions made during implementation
- Description of the shadow analysis methodology
- Description of the efficiency analysis methodology
- Explanation of any limitations or simplifications
- Details of any external libraries or frameworks used